Erfan Rahmani

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Education

Master of Science in Electrical Engineering - Control Systems 2022–2025

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

Bachelor of Science in Electrical Engineering 2017–2022

University of Isfahan, Isfahan, Iran

Experience

Graduate Research Assistant 2022–2025

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

During my MSc, I worked on the detection and mitigation of zero-dynamics anomalies in AC microgrids. On the detection side, I developed a novel switching topology-based approach for detecting stealthy attacks, which led to a paper accepted for publication in IEEE Transactions on Industrial Informatics (Q1, IF=9.8). I also developed a resilient secondary voltage controller designed to maintain reliable microgrid operation in the presence of cyberattacks.

Teaching Assistant 2024

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

As a TA for Dr. Hajar Atrianfar's Multi-Agent Systems course, I helped students understand and simulate algorithms such as consensus and containment using MATLAB/Simulink. I guided them through practical case studies involving microgrids, electric vehicles, and robotic systems, and received consistently excellent feedback throughout my tenure.

Reviewer of Conference 2024

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

Reviewed research papers submitted to the 12th Annual of International Conference on Robotics and Mechatronics (ICROM), contributing to peer-review process of the conference.

Internship 2022

National Iranian South Oil Company, Ahvaz, Iran

During a 3-month industrial internship, I actively participated in maintenance, fault diagnosis, and overhaul of three-phase induction motors. I collaborated efficiently with a team of technicians and engineers under supervision of senior electrical engineers, to effectively dismantle, inspect, rewind, and reassemble motors. This experience strengthened my practical understanding of electrical engineering, and senior engineers were all pleased with my performance.

Web Designer 2012–2017

Freelance

Before starting university, I spent much of my free time working independently as a web designer, primarily creating blogs and WordPress websites. Through these projects, I developed hands-on experience in HTML and PHP, as well as graphic and visual design.

Projects

Hierarchical AC Microgrid Control with BESSs

As the case study of my MSc thesis, I modeled a hierarchical AC microgrid integrating connected battery energy storage systems (BESSs). Using feedback linearization, the complex nonlinear dynamics of the system were transformed into a second-order representation, simplifying controller design. The resulting control architecture achieved synchronization of voltage and frequency across all distributed energy resources (DERs) to their nominal values, alongside accurate active power sharing in the absence of attacks. System robustness was further evaluated through a series of disturbance scenarios.

Switching Topology-Based Detection Scheme For Stealthy ZDAs Link to Demo

As part of my MSc thesis, I developed a novel switching topology-based scheme for detecting stealthy zero-dynamics attacks (ZDAs). The proposed approach was validated through simulation studies, stability analysis, and scalability assessment, demonstrating its effectiveness across varying system configurations.

Resilient Microgrid Voltage and Frequency Control

As another section of my MSc thesis, I developed a second-order voltage controller for AC microgrids designed to maintain resilient performance in the presence of cyberattacks. Stability analysis confirmed the controller's robust performance, while simulation results demonstrated the microgrid's resilience, achieving synchronized voltage and frequency values across all DERs alongside accurate power sharing.

Robust and Resilient Control of Multi-Agent Systems Against FDIAs Link to Demo

For my Robust Control course project, I simulated a resilient and robust distributed control protocol for multi-agent systems subject to sensor and actuator attacks. To this end, a distributed H_∞ control protocol was designed by solving a linear matrix inequality (LMI), yielding a controller capable of maintaining performance under adversarial conditions.

Single-DOF Robotic Manipulator Link to Demo

For my Industrial Internet of Things (IIoT) course project, I designed a controller by solving a linear matrix inequality (LMI), which achieved consensus among four single-link robotic arms. The proposed approach was subsequently validated through simulations conducted in MATLAB/Simulink.

Publications

E. Rahmani, H. Atrianfar, and A. Afshar, "A Novel Switching Topology-Based Detection Scheme for Stealthy ZDAs in AC Microgrids," *IEEE Transactions on Industrial Informatics*, 2026.

Skills

Programming	MATLAB/Simulink, C/C++, Python, LaTeX, HTML/Wordpress
Simulators	Atmel Studio, CodeVision, Eplan Electric, DIALux, Proteus, AutoCAD
Computer and OS	Microsoft Windows, Microsoft Office, Adobe Photoshop
Music	Electric Guitar
Languages	English, Persian

References

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